

Low-Light Image Restorer Using Zero-DCE

CSC245 Final Project

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Abstract

Low-light images often suffer from poor visibility and color distortion. We applied the Zero-DCE model to enhance dark images while preserving color and natural appearance. We retrained the model on the LOL dataset using multiple training strategies and compared the results to an unmodified baseline. Performance is evaluated using PSNR, SSIM, and human evaluation. While longer training improves quantitative metrics, human evaluators overwhelmingly preferred the 30-epoch fine-tuned model. These results show that targeted fine-tuning can significantly improve perceptual quality in low-light image enhancement.

1 Introduction

Dark photos are often low in quality and are not ideal for accurately capturing important moments. While many existing models attempt to improve image clarity, they often struggle with extreme low-light conditions and fail to preserve natural colors. This can pose an issue for facial recognition, object detection, and other vision tasks.

Recent approaches, such as Zero-DCE and its variants aim to enhance images without the usual required paired training data. These models use deep learning techniques to adjust image illumination in a data driven manner. However, sometimes models produced washed out colors, fail to preserve the fine structural details, or require extensive computational resources which limit their applicability.

We trained our Zero-DCE model, trying to tackle some of those problems. To evaluate our results, we used a combination of quantitative metrics—PSNR and SSIM—and human evaluation. Since many enhancement models tend to

produce washed-out colors, our work had a strong emphasis on preserving color accuracy while improving overall brightness and visual quality.

2 Related Works

There has been a lot of important work done on brightening and increasing clarity of photos in general. Our Zero-DCE model has several papers on it. One of them by Mi et al. (2024) proposes improvements to Zero-DCE. One of the topics it brings into light is how much the training dataset effects the final model. Because Zero-DCE requires no paired photos during training, that is to say a bright and dark photo, it can be quicker and less memory intensive. According to them it is important to have a diverse sampling of dark photos. Some should be brighter, while others dim. Currently, the curve that is computed off the exposure levels and is susceptible to fluctuations and bad performance when the training data only contains the same level of darkness. So Mi and al. include how to modify the curve calculations for a better model as well as how the training data effects the model's output. We can see how this effected our results because slightly dark images were blown out; those were made brighter than the ground truth.

Dark photo enhancement intersects other fields. One example is dark video enhancement (Lin et al., 2024). This paper discusses how dark videos can be broken down frame by frame, and brightened along with noise fixes. According to the paper, there is currently a lack of suitable training data. Dark videos may be common, but many lack a counter part suitable for training. They evaluate into video enhancement on several different models to see the benefits and cons of each. Our Zero-DCE model would be able to attempt something similar, but would have several advantages. Zero-DCE wouldn't require a bright video for training. And because it would need half the amount of videos the model would be have lower computational requirements and need less memory. This alternative approach could be a feasible solution for dark video enhancement.

3 Methods

Our project used an open source model for Low-Light-Image-Video-Enhancement, on GitHub (Kmohanku, 2025). Using the Zero-DCE model, we trained it on the LOL dataset (Rakshit, 2019), from Kaggle. The data included a total of 1000 photos. We used the training set (485 photos) for our model's training and the test set (15 photos) in the evaluation.

Compared to other research that has been done on darker photos, our dataset is quite small; many papers use over a thousand. The reason we used a smaller dataset was so that it could be run on our Mac laptops and improved time constraints.

For the experiment, we proposed two main approaches to deal with dark images. One was a naive 200 epoch model. And the other was a finely tuned

and optimized 30 epoch version. We modified the code to save checkpoints for our training and fine tuning. Model performance was evaluated using standard quantitative metrics and a human perception survey.

3.1 Training

We trained two model configurations to study the impact of training duration and fine-tuning. The first configuration is a naive model trained for 200 epochs without additional hyperparameter adjustments besides increasing the batch size of 1 to 4. The training time was a little over 8 hours total. Training for 200 epochs is commonly used, so we wanted to test it out.

The second configuration is a fine-tuned model trained for 30 epochs, designed to achieve competitive performance with reduced computational cost. Training was performed using the Adam optimizer with the standard learning rate schedule provided by the original Zero-DCE implementation. To improve training stability and image brightness in the fine-tuned model, several hyperparameters were adjusted. Specifically, the learning rate was reduced from $1e-4$ to $1e-5$, the batch size was increased from 1 to 4, and the exposure loss target mean was increased from 0.4 to 0.6. These changes were intended to encourage brighter outputs, reduce training noise, and improve overall image quality.

Due to the smaller number of training epochs, the 30-epoch model only took a couple of hours to train compared to the 200-epoch model. The loss rate of our 200 epoch model was higher than its 30 epoch counterpart. The 30 epoch model reached a final loss of approximately 0.61 at the end of its run, but the 200 epoch model reached around 0.64. The larger loss is believed to be caused by the larger learning rate in our naive run as well as the scheduler. If we made a better scheduler it might have been able to prevent the increase of accuracy loss.

During training, the most recent model checkpoints were saved. This allowed training to be paused and resumed when necessary, enabled fine-tuning from pretrained weights, and allowed selection of the best-performing model based on validation performance.

3.2 Evaluation

We evaluated our models using both objective image quality metrics and human perception. For quantitative evaluation, we report Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM), which are commonly used metrics for low-light image enhancement tasks. PSNR measures the pixel-wise reconstruction quality between the enhanced image and the ground truth, with higher values indicating lower reconstruction error. SSIM evaluates perceptual image quality by measuring structural similarity in terms of luminance, contrast, and texture, making it more aligned with human perception than PSNR.

In addition to quantitative metrics, we conducted a human perception survey to better capture visual preferences that may not be reflected by PSNR or SSIM.

Participants were shown five images for each example: the original low-light input, the ground truth image, the output of the base Zero-DCE model, the 30-epoch fine-tuned model output, and the 200-epoch model output. Survey participants were asked to evaluate visual quality based on brightness, clarity, and overall similarity to the ground truth.

We asked 3 sets of questions taking the format listed below:

- Which photo most closely resembles the one on the shown? Ground truth would be compared to the three outputs from corresponding models e.g. option 1, option 2, option 3.
- Option 1 to the original image on a scale of 1 to 10, how close is it to the image you chose? We did this for all three options

Our results from the survey and the quantitative evaluation had similar results, but there was an interesting difference in PSNR performance vs human evaluation, which we discussed more deeply in the later sections. Overall, the combination of objective and subjective evaluation provides a more comprehensive assessment of model performance, particularly for perceptual tasks such as low-light image enhancement.'

4 Experiments

The purpose of our experiments were compare two different training methods for Zero-DCE and evaluate how each model enhances low-light images. We tested a retrained model with and without fine-tuning against the unmodified Zero-DCE model. This allowed us to analyze how much improvement comes from retraining and fine-tuning on paired low-light data.

4.1 Setup

All experiments were performed using the LOL-dataset, which contains paired low-light and ground truth bright images. The dataset included:

- 485 images for training
- 15 images for testing

Only the retrained models used the LOL training data. The baseline model, provided in the original GitHub repository, was left unchanged and was not trained on LOL. It served as a reference point for baseline Zero-DCE performance.

4.2 Models Evaluated

- Baseline Zero-DCE model
 - No retraining, no fine-tuning

- 200 Epoch Retrained Model
 - Fully trained on LOL for 200 epochs
- 30 Epoch Fine-Tuned Model
 - Fine-tuned hyperparameters and retrained on LOL

4.3 Visual Models’ Comparison: Results

Table 1 shows some of the images produced by the three models. We had a total of 15 sets from the evaluation set, but chose only three to be discussed as the overall output was the same across photos. From observations the baseline is quite dark; the 30 and 200 epoch models have improved substantially. The 200 and 30 epoch models look like they are around a similar brightness, but the 200 epoch model has a more warm tint compared to the slightly more neutral composition of the 30 epoch model.

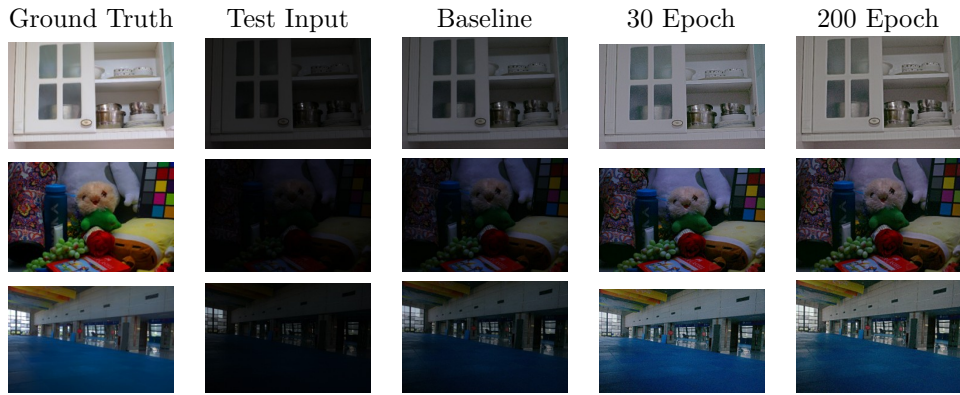


Table 1: Qualitative comparison of Zero-DCE enhancement. This contains 3 of 15 the evaluation results

4.4 Evaluation

4.4.1 Quantitative Metrics

To quantitatively compare improvement across models, we evaluated each output using PSNR and SSIM, two standard metrics for image-to-image translation tasks.

Table 2 reports the PSNR and SSIM values for each of the 15 test images across the baseline, 30-epoch, and 200-epoch models.

For each of the 15 test images, we computed these metrics and averaged the results for comparison across models. The 200-epoch model achieved the highest average PSNR, meaning it produced the cleanest pixel-level reconstructions. The 30-epoch fine-tuned model achieved the highest SSIM, indicating better

Table 2: Comparison of PSNR and SSIM Across Models

Image	Base Model		Epoch 30		Epoch 200	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
1	10.3762	0.5350	16.1568	0.6934	16.6939	0.6962
2	8.1068	0.4603	14.8822	0.6355	14.0610	0.6273
3	9.9970	0.5761	19.1017	0.7372	18.3760	0.7293
4	19.0564	0.7684	15.8403	0.6179	16.9546	0.6269
5	6.7461	0.2260	10.7708	0.4065	14.7949	0.6431
6	7.7625	0.4207	13.4387	0.6333	9.0342	0.3676
7	13.1207	0.3817	19.9037	0.5847	19.4306	0.5794
8	12.3448	0.5084	9.1421	0.3712	22.0106	0.5728
9	5.7566	0.2046	12.6590	0.3902	8.7651	0.3555
10	8.1355	0.2261	17.7529	0.5714	12.2347	0.3812
11	9.4068	0.4466	16.4324	0.5244	16.5191	0.5621
12	13.0273	0.5083	23.0606	0.4977	23.6696	0.6195
13	10.7104	0.2610	14.2788	0.6483	15.7380	0.5145
14	15.8538	0.5474	15.3873	0.7453	24.1371	0.5044
15	7.6493	0.4808	16.8768	0.4368	13.5716	0.6466
Average	10.5367	0.4368	15.6477	0.5625	16.3994	0.5618

structural and perceptual similarity. The baseline model scored the lowest on both metrics.

4.4.2 Human Evaluation

We conducted a human evaluation study to assess human perception. The participants were shown the ground truth along enhanced images from each model. Then they were asked to select the image that was most like the ground truth. Additionally, participants rated each enhanced image on a scale of 1 - 10 based on how similar they were to the ground truth image (1 being the least similar and the 10 being very similar). This evaluation allowed us to measure perceptual factors, such as color fidelity and naturalness, that are not fully reflected by numerical metrics.

Across all responses, 100% of participants selected the 30-epoch retrained model as the most similar to the original photo. The 200-epoch model was consistently rated as the second-best option, while the baseline model received the lowest ratings overall. These results indicate that fine-tuning for a shorter duration produced outputs that better aligned with human perception than longer training or the unmodified baseline.

5 Conclusions

Our results demonstrate that retraining and fine-tuning Zero-DCE on paired low-light data can significantly improve both brightness and color preservation compared to the unmodified baseline model. While longer training generally led to higher numerical performance, particularly in PSNR, these improvements did not always align with human perception of image quality.

Across both quantitative and qualitative evaluations, we observed a clear trade-off between numerical reconstruction accuracy and perceptual quality. The 200-epoch model achieved the highest average PSNR, indicating strong pixel-level reconstruction. However, human evaluators consistently preferred the 30-epoch fine-tuned model, suggesting that shorter, targeted training better preserves natural color and visual realism. This highlights an important limitation of relying solely on numerical metrics when evaluating low-light image enhancement.

Overall, our findings suggest that effective fine-tuning can produce visually superior results without requiring extensive training time or computational resources. For practical applications, especially in resource-constrained environments, shorter training with carefully chosen hyperparameters may be more effective than prolonged training. Future work could explore improved learning rate schedules, larger and more diverse datasets, or alternative perceptual metrics to further align numerical evaluation with human visual preferences.

References

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